ECON 770 Recitation 5

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Assign 2 - Question 3

Suppose that random variable X has a Laplace density $f(x) = \frac{1}{2}e^{-|x|}$ with support $\mathbb{R}$. We restrict the support on $[-K, K]$ where $K > 0$, then

1. Derive the appropriate density function $f_K(x)$ for restriction
2. Derive the moment generating function $m(\lambda) = \mathbb{E}[\exp(\lambda X)]$
3. Show $\lim_{K\to\infty} \frac{\partial}{\partial\lambda} m(\lambda)\Big|_{\lambda=0} = \mu$ where $\mu = 0$ is the mean

Assign 2 - Question 3 (Solution)

Recall the conditions of a proper density function. Obviously, $g(x) = \frac{1}{2}e^{-|x|}\mathbb{1}\{-K \leq x \leq K\}$ fits the data perfectly, but it is not a density function. So we compute

$$P(-K \leq X \leq K) = \int_0^K e^{-x}dx = 1 - e^{-K}$$

then define

$$f_K(x) = \frac{1}{2}e^{-|x|}\mathbb{1}\{-K \leq x \leq K\}(1 - e^{-K})^{-1}$$

It's clear that $\int_{-K}^{K} f_K(x)dx = 1$ and $f_K(x) \geq 0$ on its support.

Assign 2 - Question 3 (Solution, Contd)

Then we solve the moment generating function. Note that we can divide density function into two parts,

$$
\begin{aligned}
f_K(x) &= \frac{\frac{1}{2}e^{-|x|}}{1-e^{-K}}\mathbb{1}\{-K \le x \le K\} \\
&= \frac{\frac{1}{2}e^{x}}{1-e^{-K}}\mathbb{1}\{-K \le x \le 0\} + \frac{\frac{1}{2}e^{-x}}{1-e^{-K}}\mathbb{1}\{0 < x \le K\}
\end{aligned}
$$

Then,

$$
\begin{aligned}
m(\lambda) &= \mathbb{E}[e^{\lambda X}] \\
&= \int_{-K}^{K} e^{\lambda x}\left[\frac{\frac{1}{2}e^{x}}{1-e^{-K}}\mathbb{1}\{-K \le x \le 0\} + \frac{\frac{1}{2}e^{-x}}{1-e^{-K}}\mathbb{1}\{0 < x \le K\}\right]dx \\
&= \int_{-K}^{0} e^{\lambda x}\frac{\frac{1}{2}e^{x}}{1-e^{-K}}dx + \int_{0}^{K} e^{\lambda x}\frac{\frac{1}{2}e^{-x}}{1-e^{-K}}dx
\end{aligned}
$$

Assign 2 - Question 3 (Solution, Contd)

$$
\begin{aligned}
m(\lambda) &= \mathbb{E}[e^{\lambda X}] \\
&= \int_{-K}^{0} e^{\lambda x} \frac{\frac{1}{2}e^{x}}{1-e^{-K}} dx + \int_{0}^{K} e^{\lambda x} \frac{\frac{1}{2}e^{-x}}{1-e^{-K}} dx \\
&= \frac{1}{2(1-e^{-K})} \left[\int_{-K}^{0} e^{(\lambda+1)x} dx + \int_{0}^{K} e^{(\lambda-1)x} dx \right]
\end{aligned}
$$

Then we use formula $\int_0^K e^{ax} dx = \frac{1}{a}(e^{aK} - 1)$ and get

$$
m(\lambda) = \frac{1}{2(1-e^{-K})} \left[\frac{1}{\lambda - 1} \left(e^{(\lambda-1)K} - 1 \right) - \frac{1}{\lambda + 1} \left(e^{-(\lambda+1)K} - 1 \right) \right]
$$

And by algebra, $\frac{\partial}{\partial \lambda} m(\lambda)\Big|_{\lambda=0} = 0$

Assign 2 - Question 5 (Solution)

By definition, $S_Y = [0, +\infty)$ and

$$\begin{aligned} P(Y \leq y) &= P((X-\lambda)^2 \leq y) \\ &= P(\lambda - \sqrt{y} \leq X \leq \lambda + \sqrt{y}) \\ &= P(\max\{0, \lambda - \sqrt{y}\} \leq X \leq \lambda + \sqrt{y}) \end{aligned}$$

When $\sqrt{y} < \lambda$, i.e. $0 \leq y < \lambda^2$, then

$$\begin{aligned} P(Y \leq y) &= P(\lambda - \sqrt{y} \leq X \leq \lambda + \sqrt{y}) \\ &= \int_{\lambda-\sqrt{y}}^{\lambda+\sqrt{y}} \frac{1}{\lambda} e^{-x/\lambda} dx \\ &= -e^{-x/\lambda}\Big|_{\lambda-\sqrt{y}}^{\lambda+\sqrt{y}} \\ &= e^{-\frac{\lambda-\sqrt{y}}{\lambda}} - e^{-\frac{\lambda+\sqrt{y}}{\lambda}} = e^{-1}(e^{\frac{\sqrt{y}}{\lambda}} - e^{-\frac{\sqrt{y}}{\lambda}}) \end{aligned}$$

As a result, $f_Y(y) = \frac{d}{dy} P(Y \leq y) = \frac{1}{2e\lambda\sqrt{y}}(e^{\frac{\sqrt{y}}{\lambda}} + e^{-\frac{\sqrt{y}}{\lambda}})$

Assign 2 - Question 5 (Solution)

When $\sqrt{y} \geq \lambda$, i.e. $y \geq \lambda^2$, similarly we have

$$P(Y \leq y) = P(0 \leq X \leq \lambda + \sqrt{y})$$
$$= 1 - e^{-\frac{\lambda+\sqrt{y}}{\lambda}}$$

and $f_Y(y) = \frac{1}{2e\lambda\sqrt{y}} e^{-\frac{\sqrt{y}}{\lambda}}$
Therefore,

$$f_Y(y) = \mathbb{1}\{0 \leq y < \lambda^2\} \left[\frac{1}{2e\lambda\sqrt{y}} (e^{\frac{\sqrt{y}}{\lambda}} + e^{-\frac{\sqrt{y}}{\lambda}}) \right] + \mathbb{1}\{y \geq \lambda^2\} \left[\frac{1}{2e\lambda\sqrt{y}} e^{-\frac{\sqrt{y}}{\lambda}} \right]$$

Discrete - Bernoulli

- $X \sim B(\pi)$ (or written as $X \sim B(p)$)
- Support: $S_X = \{0, 1\}$
- PMF: $P(X = 1) = \pi$
- Expectation: $\mathbb{E}[X] = \pi$
- Variance: $V[X] = \pi(1 - \pi)$
- Skewness: $\tilde{\mu}_3 = \dfrac{1 - 2\pi}{\sqrt{\pi(1 - \pi)}}$

*We will use these notations (S_X, $\mathbb{E}[X]$, $V[X]$, $\tilde{\mu}_3$) later

Discrete - Binomial

- $X \sim B(\pi, n)$ (or written as $X \sim B(n, \pi)$)
- $S_X = \{0, 1, \ldots, n\}$
- $P(X = k) = \pi^k (1 - \pi)^{n-k} \dfrac{n!}{(n-k)!k!}$
- $\mathbb{E}[X] = n\pi$
- $V[X] = n\pi(1 - \pi)$

Discrete - Poisson

- $X \sim P(\lambda)$
- $P(X = k) = e^{-\lambda} \dfrac{\lambda^k}{k!}$ (and thus $\sum_{k=0}^{\infty} P(X = k) = 1$)
- $S_X = \{0, 1, 2, \dots\}$
- $\mathbb{E}[X] = \lambda$
- $V[X] = \lambda$
- If $X_i \sim P(\lambda_i)$ for $i = 1, \dots, n$ are independent, then

$$\sum_{i=1}^{n} X_i \sim P(\sum_{i=1}^{n} \lambda_i)$$

(Hint: independence)

Discrete - Binomial v.s. Poisson

Theorem

Given $X \sim B(\pi_n, n)$, if n is very big and π_n is very small s.t. $n\pi_n \to \lambda$ $(n \to \infty)$, then we have $X \sim P(\lambda)$

Proof.

Let $\pi_n = \frac{\lambda}{n}$, then

$$
\begin{aligned}
P(X = k) &= \pi_n^k (1 - \pi_n)^{n-k} \frac{n!}{(n-k)!k!} \\
&= \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \frac{n!}{(n-k)!k!} \\
&= \frac{\lambda^k}{k!} \times \frac{n!}{(n-k)!n^k} \times \frac{(1-\lambda/n)^n}{(1-\lambda/n)^k} \to \frac{\lambda^k}{k!} \times 1 \times e^{-\lambda}
\end{aligned}
$$

□

Continuous - Uniform

- $X \sim U[a,b]$
- $S_X = [a,b]$
- PDF: $f(x) = \dfrac{1}{b-a} \mathbb{1}\{x \in S_X\}$
- CDF: $F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a,b] \\ 1 & \text{for } x > b \end{cases}$
- $\mathbb{E}[X] = \dfrac{a+b}{2}$
- $V[X] = \dfrac{(b-a)^2}{12}$
- MGF: $\mathbb{E}[e^{tx}] = \begin{cases} \frac{e^{tb}-e^{ta}}{t(b-a)} & \text{for } t \neq 0 \\ 1 & \text{for } t = 0 \end{cases}$

Continuous - Exponential

- $X \sim exp(\lambda)$
- $S_X = [0, \infty)$
- PDF: $f(x; \lambda) = \frac{1}{\lambda} e^{-x/\lambda} \mathbb{1}\{x \in [0, \infty)\}$
- CDF: $F(x) = \begin{cases} 1 - e^{-x/\lambda} & x \geq 0 \\ 0 & x < 0 \end{cases}$ (exponential tails)
- $P(X > c) = e^{-c/\lambda}$
- $\mathbb{E}[X] = \lambda$
- $V[X] = \lambda^2$
- MGF: $\mathbb{E}[e^{tx}] = \dfrac{1}{1 - \lambda t}$ for $t < \dfrac{1}{\lambda}$

Continuous - Laplace

- $X \sim Laplace(\mu, b)$
- $S_X = \mathbb{R}$
- PDF: $f(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right)$
- CDF: $F(x) = \int_{-\infty}^{x} f(u)\mathrm{d}u = \begin{cases} \frac{1}{2} \exp\left(\frac{x-\mu}{b}\right) & \text{if } x < \mu \\ 1 - \frac{1}{2} \exp\left(-\frac{x-\mu}{b}\right) & \text{if } x \geq \mu \end{cases}$
- $\mathbb{E}[X] = \mu$
- $V[X] = 2b^2$
- MGF: $\mathbb{E}[e^{tx}] = \frac{\exp(\mu t)}{1-b^2t^2}$ for $|t| < 1/b$

Continuous - Pareto

- $X \sim Pareto(x_m, \alpha)$. Here let $x_m = 1$ to have the same PDF as the one in lecture slides.
- $S_X = [1, \infty)$ (in general $[x_m, \infty)$)
- PDF: $f_X(x; 1, \alpha) = \begin{cases} \frac{\alpha}{x^{\alpha+1}} & x \geq 1 \\ 0 & x < 1 \end{cases}$
- CDF: $F_X(x) = \begin{cases} 1 - \left(\frac{1}{x}\right)^{\alpha} & x \geq 1 \\ 0 & x < 1 \end{cases}$
- $\mathbb{E}[X] = \begin{cases} \infty & \text{for } \alpha \leq 1 \\ \frac{\alpha}{\alpha-1} & \text{for } \alpha > 1 \end{cases}$
- $P(X > c) = \int_c^{\infty} \frac{\alpha}{x^{\alpha+1}} dx = \int_c^{\infty} \alpha x^{-(\alpha+1)} dx = -x^{-\alpha}|_c^{\infty} = c^{-\alpha}$ which goes to 0 if c is very big, but slower[1] than exponential distribution

[1] When $X \sim exp(\lambda)$, we have $P(X > c) = e^{-c/\lambda}$

Continuous - Normal

- $X \sim N(\mu, \sigma^2)$
- $S_X = \mathbb{R}$
- PDF: $f_X(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(x-\mu)^2)$
- CDF[2]: $F_X(x) = \int_{-\infty}^{x} f_X(t)dt$
- $\mathbb{E}[X] = \mu$
- $V[X] = \sigma^2$
- $\tilde{\mu}_3$: 0
- Kurtosis $\tilde{\mu}_4$: 3
- MGF: $\mathbb{E}[e^{tX}] = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$

[2] The integral cannot be expressed in terms of elementary functions

Continuous - Normal (Moments)

Recall that we have

$$\mathbb{E}[X^n] = M_X^{(n)}(0) = \left.\frac{d^n M_X}{dt^n}\right|_{t=0}$$

so given $M_X(t) = \mathbb{E}[e^{tX}] = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$,

Order	$\mathbb{E}[X^n]$	$\mathbb{E}[(X-\mu)^n]$
1	μ	0
2	$\mu^2 + \sigma^2$	σ^2
3	$\mu^3 + 3\mu\sigma^2$	0
4	$\mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$	$3\sigma^4$
5	$\mu^5 + 10\mu^3\sigma^2 + 15\mu\sigma^4$	0
...		

Continuous - Normal (Properties)

- "Standardize": Given $X \sim N(\mu, \sigma^2)$, let $Y = \dfrac{X-\mu}{\sigma}$ and then

$$Y \sim N(0,1)$$

  On the other hand, if $Z = \dfrac{Y-a}{b} \sim N(\mu, \sigma^2)$, then $Y \sim N(a + b\mu, b^2\sigma^2)$
- Closed under affine combination

$$Y = b + \sum_{i=1}^{n} a_i X_i \sim N(b + \sum_{i=1}^{n} a_i \mu_i, \sum_{i=1}^{n} a_i^2 \sigma_i^2)$$

  where $X_i \sim N(\mu_i, \sigma_i^2)$ are mutually independent

Proof.

By using MGF

$$\mathbb{E}[e^{tX_1+tX_2+\cdots+tX_n}] = \mathbb{E}[e^{tX_1}]\mathbb{E}[e^{tX_2}]\ldots\mathbb{E}[e^{tX_n}] = exp\left(t\sum_{i=1}^{n}\mu_i + \frac{1}{2}t^2\sum_{i=1}^{n}\sigma_i^2\right)$$

Continuous - Log Normal

- X is log-normal if $\ln(X) \sim N(\mu, \sigma^2)$ $(X > 0)$
- $S_X = (0, \infty)$
- PDF: $f_X(x) = \dfrac{1}{x\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(\ln(x) - \mu)^2)$
- CDF[3]: $F_X(x) = \int_{-\infty}^{x} f_X(t)dt$
- $\mathbb{E}[X] = \exp(\mu + \dfrac{\sigma^2}{2})$
- $V[X] = \left(\exp(\sigma^2) - 1\right) \exp(2\mu + \sigma^2)$
- Median: $\exp(\mu)$ (by FOC)

[3] The integral cannot be expressed in terms of elementary functions

Continuous - Chi-Squared

- If $\{X_i\}_{i=1}^k$ are i.i.d. $N(0,1)$, then $Y = \sum_{i=1}^k X_i^2 \sim \mathcal{X}^2(k)$ is Chi-squared distributed with k degrees of freedom.
- $S_X = (0, +\infty)$ if $k = 1$, otherwise $S_X = [0, +\infty)$
- PDF:

$$f_X(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$$

  where $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$ (Gamma function)
- $\mathbb{E}[X] = k$
- $V[X] = 2k$
- If $\{Y_i\}_{i=1}^n$ are independently Chi-squared distributed with $\{k_i\}_{i=1}^n$ degrees of freedom, then

$$Z \equiv \sum_{i=1}^n Y_i \sim \mathcal{X}^2(\sum_{i=1}^n k_i)$$